6.1 Unit description

Introduction

This unit contains a practical written examination that covers the content of Units 4 and 5. There is no specific content for this unit.

Development of practical skills, knowledge and understanding

Students are expected to develop experimental skills, and a knowledge and understanding of the necessary techniques, by carrying out a range of practicals while they study Units 4 and 5.

This unit will assess students' knowledge and understanding of practical procedures and techniques that they develop.

To prepare them for the assessment of this unit centres should provide opportunities for students to carry out practical activities, collect and analyse data, and draw conclusions.

By completing these practicals students will be able to:

- follow and interpret experimental instructions, covering the full range of laboratory exercises set throughout the course, with minimal help from the teacher
- always work with interest and enthusiasm in the laboratory completing most laboratory exercises in the time allocated
- manipulate apparatus, use chemicals, carry out all common laboratory procedures and use data logging (where appropriate) with the highest level of skill that may be reasonably expected at this level
- work sensibly and safely in the laboratory paying due regard to health and safety requirements without the need for reminders from the teacher
- gain accurate and consistent results in quantitative exercises, make most of the expected observations in qualitative exercises and obtain products in preparations of high yield and purity.

These practical activities should cover a range of different topic areas and require the use of a variety of practical techniques.

How chemists work

Students should be given the opportunity to develop their practical skills for *How Science Works* by completing a range of different practicals that require a variety of different techniques.

Students' laboratory reports on their practical work should use appropriate scientific, technical and mathematical language, conventions and symbols in order to meet the requirements of *How Science Works*.

6.2 Assessment information

Unit 6 examination

This unit contains a practical written examination that covers the content of Units 4 and 5.

The practical written examination covers the following types of practicals:

- qualitative observations
- quantitative measurements
- preparations.

The examination will last 1 hour 15 minutes and have 50 marks. It will contain one section.

Students may use a calculator.

The quality of written communication will be assessed in the practical written examination. When answering these questions students will be assessed on their ability to organise and present information, ideas, descriptions and arguments clearly and logically, including the use of grammar, punctuation and spelling.

D Assessment and additional information

Assessment information

Assessment requirements

For a summary of assessment requirements and assessment objectives, see *Section B: Specification overview*.

Entering candidates for the examinations for this qualification

Details of how to enter candidates for the examinations for this qualification can be found in the International Information Manual, copies of which are sent to all examinations officers. The information can also be found at: www.edexcel.com/international

Resitting of units

There is one resit opportunity allowed for each unit prior to claiming certification for the qualification. The best available result for each contributing unit will count towards the final grade.

After certification all unit results may be reused to count towards a new award. Students may re-enter for certification only if they have retaken at least one unit.

Results of units are held in the Pearson unit bank and have a shelf life limited only by the shelf life of this specification.

Awarding and reporting

The IAS qualification will be graded and certificated on a five-grade scale from A to E. The full International Advanced Level will be graded on a six-point scale A* to E. Individual unit results will be reported.

A pass in an International Advanced Subsidiary subject is indicated by one of the five grades A, B, C, D, E of which grade A is the highest and grade E the lowest. A pass in an International Advanced Level subject is indicated by one of the six grades A*, A, B, C, D, E of which grade A* is the highest and grade E the lowest. To be awarded an A* students will need to achieve an A on the full International Advanced Level qualification and an A* aggregate of the IA2 units. Students whose level of achievement is below the minimum judged by Pearson to be of sufficient standard to be recorded on a certificate will receive an unclassified U result.

Performance descriptions

Performance descriptions give the minimum acceptable level for a grade. See *Appendix 1* for the performance descriptions for this subject.

Unit results

The minimum uniform marks required for each grade for each unit:

Unit 1

Unit grade	A	B	C	D	E
Maximum uniform mark = 120	**96**	**84**	**72**	**60**	**48**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–47.

Unit 2

Unit grade	A	B	C	D	E
Maximum uniform mark = 120	**96**	**84**	**72**	**60**	**48**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–47.

Unit 3

Unit grade	A	B	C	D	E
Maximum uniform mark = 60	**48**	**42**	**36**	**30**	**24**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–23.

Unit 4

Unit grade	A	B	C	D	E
Maximum uniform mark = 120	**96**	**84**	**72**	**60**	**48**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–47.

Unit 5

Unit grade	A	B	C	D	E
Maximum uniform mark = 120	**96**	**84**	**72**	**60**	**48**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–47.

Unit 6

Unit grade	A	B	C	D	E
Maximum uniform mark = 60	**48**	**42**	**36**	**30**	**24**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–23.

Qualification results

The minimum uniform marks required for each grade:

International Advanced Subsidiary cash-in code XCH01

Qualification grade	A	B	C	D	E
Maximum uniform mark = 300	**240**	**210**	**180**	**150**	**120**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–119.

International Advanced Level cash-in code YCH01

Qualification grade	A	B	C	D	E
Maximum uniform mark = 600	**480**	**420**	**360**	**300**	**240**

Students who do not achieve the standard required for a grade E will receive a uniform mark in the range 0–239.

To be awarded an A* students will need to achieve an A on the full International Advanced Level qualification and an A* aggregate of the IA2 units.

Language of assessment	Assessment of this specification will be available in English only. Assessment materials will be published in English only and all work submitted for examination must be produced in English.
Quality of written communication	Students will be assessed on their ability to: ■ write legibly, with accurate use of spelling, grammar and punctuation in order to make the meaning clear ■ organise relevant information clearly and coherently, using specialist vocabulary when appropriate. In IAL Chemistry the quality of written communication will cover all of these, with students selecting the most relevant way of communicating their information, to a particular context. Quality of written communication will be assessed in all units.
Synoptic assessment	In synoptic assessment there should be a concentration on the quality of assessment to ensure that it encourages the development of the holestic understanding of the subject. Synopticity requires students to connect knowledge, understanding and skills acquired in different parts of the International Advanced Level course. Synoptic assessment in the context of chemistry requires students to apply knowledge and understanding gained from other units to solve a problem. For example, problems related to organic chemistry in Unit 5 require all of the knowledge and understanding the students have developed throughout the other IAS and IA2 topics. In the Unit 4 and 5 examinations at IA2 there will be three sections. Sections B and C will contain extended answer questions where students can demonstrate the scientific knowledge they have developed over the whole IAL in Chemistry. These sections will address the synoptic assessment of the IAL in Chemistry. In addition there will be specific questions in Section C of the examinations which will address synopticity. In Unit 4 students will have to answer data questions, using knowledge and understanding gained from the IAS units and Unit 4. In Unit 5 students will have to answer contextualised questions on the areas of Unit 5, drawing on their knowledge and understanding gained throughout the whole IAL in Chemistry.

Additional information

Malpractice

For up-to-date information on malpractice, please refer to the latest Joint Council for Qualifications (JCQ) Suspected Malpractice in Examinations and Assessments: Policies and Procedures document, available on the JCQ website: www.jcq.org.uk

Access arrangements and special requirements

Pearson's policy on access arrangements and special considerations for GCE, GCSE, IAL and Entry Level is designed to ensure equal access to qualifications for all students (in compliance with the Equality Act 2010) without compromising the assessment of skills, knowledge, understanding or competence.

Please see the JCQ website (www.jcq.org.uk) for their policy on access arrangements, reasonable adjustments and special considerations.

Please see our website (www.edexcel.com) for:

- the forms to submit for requests for access arrangements and special considerations
- dates for submissions of the forms.

Equality Act 2010

Please see our website (www.edexcel.com) for information on the Equality Act 2010.

Prior learning and progression

Prior learning

Students who would benefit most from studying an International Advanced Level in Chemistry are likely to have a Level 2 qualification such as an International GCSE in Chemistry at grades A* – C.

Progression

This qualification supports progression into further education, training or employment.

Combinations of entry

Only units achieved from this qualification may contribute to the certification of the International Advanced Subsidiary in Chemistry or the International Advanced Level in Chemistry.

Student recruitment

Pearson's access policy concerning recruitment to our qualifications is that:

- they must be available to anyone who is capable of reaching the required standard

E Support, training and resources

Support

Pearson aim to provide the most comprehensive support for our qualifications. Here are just a few of the support services we offer:

- Subject Advisor – subject experts are on-hand to offer their expertise to answer any questions you may have on delivering the qualification and assessment.
- Subject Page – written by our Subject Advisors, the subject pages keep you up to date with the latest information on your subject.
- Subject Communities – exchange views and share information about your subject with other teachers.
- Training – see 'Training' below for full details.

For full details of all the teacher and student support provided by Pearson to help you deliver our qualifications, please visit:
www.edexcel.com/ial/chemistry/support

Training

Our programme of professional development and training courses, covering various aspects of the specification and examinations, are arranged each year on a regional basis. Pearson training is designed to fit you, with an option of face-to-face, online or customised training so you can choose where, when and how you want to be trained.

Face-to-face training
Our programmes of face-to-face training have been designed to help anyone who is interested in, or currently teaching, a Pearson Edexcel qualification. We run a schedule of events throughout the academic year to support you and help you to deliver our qualifications.

Online training
Online training is available for international centres who are interested in, or currently delivering, our qualifications. This delivery method helps us run training courses more frequently to a wider audience.

To find out more information or to book a place please visit:
www.edexcel.com/training

Alternatively, email internationaltfp@pearson.com or
telephone +44 (0) 44 844 576 0025

Resources

Pearson is committed to ensuring that teachers and students have a choice of resources to support their teaching and study.

Teachers and students can continue to use their existing GCE A level resources for International Advanced Levels.

To search for Pearson GCE resources, please visit: www.pearsonschools.co.uk

To search for endorsed resources from other publishers, please visit: www.edexcel.com/resources

Specifications, Sample Assessment Materials and Teacher Support Materials

Specifications, Sample Assessment Materials (SAMs) and Teacher Support Materials (TSMs) can be downloaded from the International Advanced Level subject pages.

To find a complete list of supporting documents, including the specification, SAMs and TSMs, please visit: www.edexcel.com/ial/chemistry

F Appendices

Appendix 1 Performance descriptions

Introduction

Performance descriptions describe the learning outcomes and levels of attainment likely to be demonstrated by a representative candidate performing at the A/B and E/U boundaries for IAS and IA2.

In practice most candidates will show uneven profiles across the attainments listed, with strengths in some areas compensating in the award process for weaknesses or omissions elsewhere. Performance descriptions illustrate expectations at the A/B and E/U boundaries of the IAS and IA2 as a whole; they have not been written at unit level.

Grade A/B and E/U boundaries should be set using professional judgement. The judgement should reflect the quality of candidates' work, informed by the available technical and statistical evidence. Performance descriptions are designed to assist examiners in exercising their professional judgement. They should be interpreted and applied in the context of individual specifications and their associated units. However, performance descriptions are not designed to define the content of specifications and units.

IAS performance descriptors for Chemistry

	Assessment objective 1	Assessment objective 2	Assessment objective 3
t	**Knowledge and understanding of science and of *How Science Works*** Candidates should be able to: ■ recognise, recall and show understanding of scientific knowledge ■ select, organise and communicate relevant information in a variety of forms.	**Application of knowledge and understanding of science and of *How Science Works*** Candidates should be able to: ■ analyse and evaluate scientific knowledge and processes ■ apply scientific knowledge and processes to unfamiliar situations including those related to issues ■ assess the validity, reliability and credibility of scientific information.	***How Science Works*** Candidates should be able to: ■ demonstrate and describe ethical, safe and skilful practical techniques and processes, selecting appropriate qualitative and quantitative methods ■ make, record and communicate reliable and valid observations and measurements with appropriate precision and accuracy ■ analyse, interpret, explain and evaluate the methodology, results and impact of their own and others' experimental and investigative activities in a variety of ways.
dary ce ns	Candidates characteristically: a) demonstrate knowledge and understanding of most principles, concepts and facts from the IAS specification b) select relevant information from the IAS specification c) organise and present information clearly in appropriate forms using scientific terminology d) write equations for most straightforward reactions using scientific terminology.	Candidates characteristically: a) apply principles and concepts in familiar and new contexts involving only a few steps in the argument b) describe significant trends and patterns shown by data presented in tabular or graphical form; interpret phenomena with few errors; and present arguments and evaluations clearly c) comment critically on statements, conclusions or data d) carry out accurately most structured calculations specified for IAS e) use a range of chemical equations f) translate successfully data presented as prose, diagrams, drawings, tables or graphs from one form to another.	Candidates characteristically: a) devise and plan experimental and investigative activities, selecting appropriate techniques b) demonstrate safe and skilful practical techniques c) make observations and measurements with appropriate precision and record these methodically d) interpret, explain, evaluate and communicate the results of their own and others' experimental and investigative activities, in appropriate contexts.

	Assessment objective 1	Assessment objective 2	Assessment objective 3
E/U boundary performance descriptions	Candidates characteristically: a) demonstrate knowledge and understanding of some principles and facts from the IAS specification b) select some relevant information from the IAS specification c) present information using basic terminology from the IAS specification d) write equations for some straightforward reactions.	Candidates characteristically: a) apply a given principle to material presented in familiar or closely related contexts involving only a few steps in the argument b) describe some trends or patterns shown by data presented in tabular or graphical form c) identify, when directed, inconsistencies in conclusions or data d) carry out some steps within calculations e) use simple chemical equations f) translate data successfully from one form to another, in some contexts.	Candidates characteristically: a) devise and plan some aspects of experimental and investigative activities b) demonstrate safe practical techniques c) make observations and measurements and record them d) interpret, explain and communicate some aspects of the results of their own and others' experimental and investigative activities, in appropriate contexts.

IA2 performance descriptors for Chemistry

	Assessment objective 1	Assessment objective 2	Assessment objective 3
t	**Knowledge and understanding of science and of *How Science Works*** Candidates should be able to: ■ recognise, recall and show understanding of scientific knowledge ■ select, organise and communicate relevant information in a variety of forms.	**Application of knowledge and understanding of science and of *How Science Works*** Candidates should be able to: ■ analyse and evaluate scientific knowledge and processes ■ apply scientific knowledge and processes to unfamiliar situations including those related to issues ■ assess the validity, reliability and credibility of scientific information.	***How Science Works*** Candidates should be able to: ■ demonstrate and describe ethical, safe and skilful practical techniques and processes, selecting appropriate qualitative and quantitative methods ■ make, record and communicate reliable and valid observations and measurements with appropriate precision and accuracy ■ analyse, interpret, explain and evaluate the methodology, results and impact of their own and others' experimental and investigative activities in a variety of ways.
dary ce ns	Candidates characteristically: a) demonstrate detailed knowledge and understanding of most principles, concepts and facts from the IA2 specification b) select relevant information from the IA2 specification c) organise and present information clearly in appropriate forms using scientific terminology d) write equations for most chemical reactions.	Candidates characteristically: a) apply principles and concepts in familiar and new contexts involving several steps in the argument b) describe significant trends and patterns shown by complex data presented in tabular or graphical form; interpret phenomena with few errors; and present arguments and evaluations clearly c) evaluate critically any statements, conclusions or data d) carry out accurately complex calculations specified for International Advanced level e) use chemical equations in a range of contexts f) translate successfully data presented as prose, diagrams, drawings, tables or graphs from one form to another g) select a wide range of facts, principles and concepts from both IAS and IA2 specifications h) link together appropriate facts principles and concepts from different areas of the specification.	Candidates characteristically: a) devise and plan experimental and investigative activities, selecting appropriate techniques b) demonstrate safe and skilful practical techniques c) make observations and measurements with appropriate precision and record these methodically d) interpret, explain, evaluate and communicate the results of their own and others' experimental and investigative activities, in appropriate contexts.

	Assessment objective 1	Assessment objective 2	Assessment objective 3
E/U boundary performance descriptions	Candidates characteristically: a) demonstrate knowledge and understanding of some principles, concepts and facts from the IA2 specification b) select some relevant information from the IA2 specification c) present information using basic terminology from the IA2 specification d) write equations for some chemical reactions.	Candidates characteristically: a) apply given principles or concepts in familiar and new contexts involving a few steps in the argument b) describe, and provide a limited explanation of, trends or patterns shown by complex data presented in tabular or graphical form c) identify, when directed, inconsistencies in conclusions or data d) carry out some steps within calculations e) use some chemical equations f) translate data successfully from one form to another, in some contexts g) select some facts, principles and concepts from both IAS and IA2 specifications h) put together some facts, principles and concepts from different areas of the specification.	Candidates characteristically: a) devise and plan some aspects of experimental and investigative activities b) demonstrate safe practical techniques c) make observations and measurements and record them d) interpret, explain and communicate some of the results of their own and others' experimental and investigative activities, in appropriate contexts.

Appendix 2 Codes

Type of code	Use of code	Code number
Unit codes	Each unit is assigned a unit code. This unit code is used as an entry code to indicate that a student wishes to take the assessment for that unit. Centres will need to use the entry codes only when entering students for their examination.	Unit 1 – WCH01 Unit 2 – WCH02 Unit 3 – WCH03 Unit 4 – WCH04 Unit 5 – WCH05 Unit 6 – WCH06
Cash-in codes	The cash-in code is used as an entry code to aggregate the student's unit scores to obtain the overall grade for the qualification. Centres will need to use the entry codes only when entering students for their qualification.	IAS – XCH01 IAL – YCH01
Entry codes	The entry codes are used to: 1 enter a student for the assessment of a unit 2 aggregate the student's unit scores to obtain the overall grade for the qualification.	Please refer to the *Pearson Information Manual*, available on our website (www.edexcel.com).

Appendix 3 How Science Works — mapping and expansion on specification content

How Science Works — mapping

How Science Works reference	Specification reference			
	Unit 1	Unit 2	Unit 4	Unit 5
1 Use theories, models and ideas to develop and modify scientific explanations	1.4d 1.4e 1.4g 1.5e 1.5f 1.5g 1.5k 1.6.1a 1.6.1h 1.6.1l 1.6.3b 1.7.2e 1.7.3b 1.7.3e	2.3c 2.4a 2.4b 2.4c 2.4d 2.5b 2.5d 2.6a all of 2.7 2.8b 2.8c 2.8d 2.8e 2.11a 2.11b 2.11c 2.11d 2.11e 2.11f	4.3h 4.3j all of 4.4 4.5c 4.5f 4.5g 4.5h all of 4.6 4.7a 4.7b 4.7c 4.7e	5.3.1c 5.3.1d 5.3.1e 5.3.1f 5.3.2c 5.4.1a
2 Use knowledge and understanding to pose scientific questions, define scientific problems, and present scientific arguments and ideas	1.7.2c 1.7.2d 1.7.2e 1.7.3b 1.7.3h	2.11g 2.12d 2.13b	4.6a 4.6b 4.6c 4.6d 4.7g	5.3.2g 5.4.2h 5.4.3a 5.4.3d
3 Use appropriate methodology, including ICT, to answer scientific questions and solve scientific problems	1.4d 1.4f 1.5b 1.7.2c 1.7.2d 1.7.3h	2.11g	4.3c 4.3d 4.3e 4.8.1d	5.4.3a 5.4.3d
4 Carry out experimental and investigative activities, including appropriate risk management, in a range of contexts	General laboratory work			

How Science Works reference	Specification reference			
	Unit 1	Unit 2	Unit 4	Unit 5
5 Analyse and interpret data to provide evidence, recognising correlations and causal relationships	1.3j 1.4f 1.7.1c	all of 2.7 2.11g	2.7 2.11g 4.5c 4.8.1d	5.3.1l 5.4.3b 5.4.3c
6 Evaluate methodology, evidence and data and resolve conflicting evidence	1.3i 1.7.3h			1.3i 1.7.3h
7 Appreciate the tentative nature of scientific knowledge	1.5e 1.5f	2.4a 2.4b 2.4c 2.4d		5.3.1d 5.3.1e 5.3.1f 5.3.2c 5.4.1a
8 Communicate information and ideas in appropriate ways using appropriate terminology	1.7.2c 1.7.2d 1.7.3h	2.11g all of 2.13	4.6a 4.6b 4.6c 4.6d	5.3.1i 5.3.2h
9 Consider the applications and implications of science and appreciate their associated benefits and risks	1.7.1c 1.7.1d 1.7.3d 1.7.3h	2.3e 2.9c 2.10.2f all of 2.13	4.6a 4.6b 4.6c 4.6d 4.7m all of 4.9	5.3.1j 5.3.1k 5.3.2h 5.3.2k 5.4.3a 5.4.3d
10 Consider ethical issues in the treatment of humans, other organisms and the environment		2.3e 2.11g all of 2.13	4.6a 4.6b 4.6c 4.6d all of 4.9	5.3.2l
11 Appreciate the role of the scientific community in validating new knowledge and ensuring integrity	1.7.3h	all of 2.13	all of 4.9	5.3.2h 5.4.3b 5.4.3c
12 Appreciate the ways in which society uses science to inform decision making	1.3c 1.7.2c 1.7.2d 1.7.3h	2.7.1h 2.11g all of 2.13	4.6a 4.6b 4.6c 4.6d all of 4.9	5.3.1i